moon.jpg file exists. The file exists, so the middleware retrieves it and returns it as the response to the web server and, ultimately, to the browser.

If the file doesn't exist, the request effectively passes *through* the static-file middleware unchanged. But wait—you added only one piece of middleware, right? Surely you can't pass the request through to the next middleware component if there *isn't* another one.

ASP.NET Core automatically adds a dummy piece of middleware to the end of the pipeline. This middleware always returns a 404 response if it's called.

TIP If no middleware generates a response for a request, the pipeline automatically returns a simple 404 error response to the browser.

HTTP response status codes

Every HTTP response contains a *status code* and, optionally, a *reason phrase* describing the status code. Status codes are fundamental to the HTTP protocol and are a standardized way of indicating common results. A 200 response, for example, means that the request was successfully answered, whereas a 404 response indicates that the resource requested couldn't be found. You can see the full list of standardized status codes at https://www.rfc-editor.org/rfc/rfc9110#name-status-codes.

Status codes are always three digits long and are grouped in five classes, based on the first digit:

- *1xx*—Information. This code is not often used; it provides a general acknowledgment.
- *2xx*—Success. The request was successfully handled and processed.
- *3xx*—Redirection. The browser must follow the provided link to allow the user to log in, for example.
- *4xx*—Client error. A problem occurred with the request. The request sent invalid data, for example, or the user isn't authorized to perform the request.